CLMS to Copernicus Data Space Ecosystem Migration

Weekly Status Update

Land Monitoring Service

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Date: **2026-07-16**
Version: **1.0.0**

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```
// ── Load data from external URL (not from repo) ──
data = await fetch("https://raw.githubusercontent.com/copernicus-land/clms-cdse-migration-data/main/migration_data.json")
  .then(r => r.json())
  .catch(e => { console.error("Failed to load data", e); return {groups:[], portfolio:{}, cdse:{}, summary:{}} })
groups = data.groups
summary = data.summary
portfolio = data.portfolio
cdse = data.cdse
lastUpdated = data.scraped_at.slice(0,10)

// ── Filter groups ──
filtered = groups.filter(d => {
  if (!activeComponents.has(d.component)) return false
  if (searchTerm && !d.name.toLowerCase().includes(searchTerm.toLowerCase())) return false
  return true
})

// ── Aggregate by category ──
categories = [...new Set(filtered.map(d => d.category).filter(Boolean))].sort()

// ── Totals reflecting the current filter selection ──
filteredTotals = ({
  datasets: d3.sum(filtered, d => d.total),
  groups: filtered.length
})
catStats = categories.map(cat => {
  const g = filtered.filter(d => d.category === cat)
  return {
    name: cat,
    groups: g,
    total: d3.sum(g, d => d.total),
    on_cdse: d3.sum(g, d => d.on_cdse),
    pct: Math.round(d3.sum(g, d => d.on_cdse) / Math.max(d3.sum(g, d => d.total), 1) * 100),
    s3: d3.sum(g, d => d.s3),
    stac: d3.sum(g, d => d.stac),
    openeo: d3.sum(g, d => d.openeo),
    shub: d3.sum(g, d => d.shub),
  }
})

// ── Helpers ──
pctColor = (pct) => pct >= 100 ? "#00aa00" : pct >= 75 ? "#66cc00" : pct >= 50 ? "#ffcc00" : pct >= 25 ? "#ff8800" : "#ff0000"

pctBar = (pct) => {
  const capped = Math.min(pct, 100)
  const color = pctColor(pct)
  const filled = "█".repeat(Math.floor(capped / 20))
  const empty = "░".repeat(Math.max(0, 5 - Math.floor(capped / 20)))
  return htl.html`<span style="color:${color};font-weight:bold">${pct}%</span> <span style="color:#bbb">${filled}${empty}</span>`
}
```

```
badge = (val, total) => {
  if (val === 0 && total === 0) return htl.html`<span style="background:#888;color:#fff;padding:1px 6px;border-radius:3px;font-weight:bold">—</span>`
  const pct = Math.min(val / Math.max(total, 1), 1)
  const color = pct >= 1 ? "#00aa00" : pct >= 0.75 ? "#66cc00" : pct >= 0.5 ? "#ffcc00" : pct >= 0.25 ? "#ff8800" : "#ff0000"
  return htl.html`<span style="background:${color};color:#fff;padding:1px 6px;border-radius:3px;font-weight:bold">${val}</span>`
}

check = (on) => on
  ? htl.html`<span style="color:#00aa00;font-weight:bold">✓</span>`
  : htl.html`<span style="color:#ccc">✗</span>`

// ── Changes since last week ──
changes = data.changes_since_last_week || []
newGroups = changes.filter(d => d.newly_on_cdse)
positiveChanges = changes.filter(d => d.s3 > 0 && !d.newly_on_cdse)
lastWeek = changes.length > 0 ? "since " + data.scraped_at.slice(0,10) : ""
odataDelta = data.odata_delta || 0

// ── Toggle functions for expand/collapse (plain DOM, no OJS reactivity) ──
toggleScript = {
  window.toggleCategory = function(catName) {
    const key = catName.replace(/"/g, '&quot;');
    const groupRows = document.querySelectorAll('tr[data-category="' + key + '"]:not([data-group])');
    const isExpanded = groupRows.length > 0 && groupRows[0].style.display !== 'none';
    groupRows.forEach(function(r) { r.style.display = isExpanded ? 'none' : ''; });
    if (isExpanded) {
      // Collapsing the category also collapses any open dataset rows within it
      document.querySelectorAll('tr[data-category="' + key + '"][data-group]').forEach(function(r) { r.style.display = 'none'; });
      document.querySelectorAll('span[data-group-arrow^="' + key + '::"]').forEach(function(a) { a.textContent = '▶'; });
    }
    const arrow = document.querySelector('span[data-arrow="' + key + '"]');
    if (arrow) arrow.textContent = isExpanded ? '▶' : '▼';
  }
  window.toggleGroup = function(groupKey) {
    const key = groupKey.replace(/"/g, '&quot;');
    const rows = document.querySelectorAll('tr[data-group="' + key + '"]');
    const isExpanded = rows.length > 0 && rows[0].style.display !== 'none';
    rows.forEach(function(r) { r.style.display = isExpanded ? 'none' : ''; });
    const arrow = document.querySelector('span[data-group-arrow="' + key + '"]');
    if (arrow) arrow.textContent = isExpanded ? '▶' : '▼';
  }
}
```

1. Changes since last week

```
html`
${newGroups.length > 0 ? htl.html`
<div style="background:#e8f5e9;border:1px solid #a5d6a7;border-radius:6px;padding:12px;margin:12px 0">
  <strong>□ Newly onboarded CLMS products ${lastWeek}:</strong> ${newGroups.map(d => d.name).join(", ")}
</div>` : ""}
${positiveChanges.length > 0 ? htl.html`
<div style="background:#e3f2fd;border:1px solid #90caf9;border-radius:6px;padding:12px 12px 4px 12px;margin:12px 0">
   <strong>□ More data ingested:</strong>
    ${positiveChanges.map(d => htl.html`<div style="margin:4px 0">${d.name}: <strong>+${d.s3}</strong> S3, <strong>+${d.stac}</strong> STAC, <strong>+${d.openeo}</strong> OpenEO</div>`)}
</div>` : ""}
${changes.length === 0 ? htl.html`
<div style="background:#f5f5f5;border:1px solid #ddd;border-radius:6px;padding:12px;margin:12px 0;color:#888">No new CLMS products were onboarded since last week.</div>` : ""}
${odataDelta !== 0 ? htl.html`
<div style="background:#fff3e0;border:1px solid #ffcc80;border-radius:6px;padding:12px;margin:12px 0">
    <strong>📦 New CLMS files ingested since last week:</strong> ${odataDelta > 0 ? "+" : ""}${odataDelta.toLocaleString()}
</div>` : ""}
`
```

2. Status Overview

The CLMS portfolio is composed of `{ojs} filteredTotals.datasets` datasets in `{ojs} filteredTotals.groups` product groups (current selection). Of these, the following are available on the CDSE:

Metric	Value
CLMS in S3	`{ojs} cdse.total_dataset_ids` dataset IDs
CLMS via OData	`{ojs} cdse.total_products_odata.toLocaleString()` `{ojs} odataDelta > 0 ? "(+" + odataDelta.toLocaleString() + ")" : odataDelta < 0 ? "(" + odataDelta.toLocaleString() + ")" : ""` files
CLMS via STAC	`{ojs} cdse.total_stac_collections` collections
CLMS via OpenEO	`{ojs} cdse.total_openeo_collections` collections

3. CLMS Product on CDSE

```
// Toggle-button style filter: click a component to enable/disable it
// (multi-select, all enabled by default). Replaces the old single-select
// dropdown, which only let one component be viewed at a time.
viewof activeComponents = {
```

```
  const allComponents = [...new Set(groups.map(d => d.component).filter(Boolean))].sort()
  const active = new Set(allComponents)

  const container = document.createElement("div")
     container.style.cssText = "display:flex;gap:8px;flex-wrap:wrap;align-items:center;margin-bottom:12px";

  function render() {
    container.innerHTML = "";
    allComponents.forEach(comp => {
      const isActive = active.has(comp);
      const btn = document.createElement("button");
      btn.type = "button";
      btn.textContent = comp;
               btn.style.cssText = `padding:6px 14px;border-radius:16px;cursor:pointer;font-size:0.85em;border:1px solid ${isActive ? "#0d6efd" : "#ccc"};background:${isActive ? "#0d6efd" : "#f5f5f5"};color:${isActive ? "#fff" : "#555"}`;
      btn.addEventListener("click", () => {
        if (active.has(comp)) active.delete(comp);
        else active.add(comp);
        render();
        container.value = new Set(active);
        container.dispatchEvent(new CustomEvent("input"));
      });
      container.appendChild(btn);
    });
  }
  render();
  container.value = new Set(active);
  return container;
}

viewof searchTerm = Inputs.text({label: "Search", placeholder: "Product group…", width: 250})
```

```
// ── Build the table via plain DOM APIs ──
// so that column widths defined on <colgroup> are guaranteed to apply to
// every row, including rows toggled between hidden/visible.
productGroupsTable = {
  const COLS = 8;
  const colWidths = ["auto", "9%", "9%", "9%", "9%", "9%", "9%", "18%"];

  function makeCell(tag, html, opts = {}) {
    const el = document.createElement(tag);
    if (html instanceof Node) el.appendChild(html);
    else el.innerHTML = html;
    el.style.padding = opts.padding ?? "6px";
    el.style.whiteSpace = "nowrap";
    el.style.overflow = "hidden";
    el.style.textOverflow = "ellipsis";
    if (opts.center) el.style.textAlign = "center";
    if (opts.bold) el.style.fontWeight = "bold";
    if (opts.title) el.title = opts.title;
    return el;
  }
```

```
  const wrap = document.createElement("div");

  const table = document.createElement("table");
  table.style.cssText = "width:100%;border-collapse:collapse;font-size:0.8em;table-layout:fixed";

  const colgroup = document.createElement("colgroup");
  colWidths.forEach(w => {
    const col = document.createElement("col");
    col.style.width = w;
    colgroup.appendChild(col);
  });
  table.appendChild(colgroup);

  const thead = document.createElement("thead");
  const headRow = document.createElement("tr");
  headRow.style.cssText = "background:#f0f0f0;text-align:left";
  ["Category", "Total", "CDSE", "S3", "STAC", "OpenEO", "S.Hub", "Progress"].forEach((label, i)
=> {
    const th = makeCell("th", label, { center: i > 0 && i < 7, padding: "6px" });
    th.style.borderBottom = "2px solid #ddd";
    headRow.appendChild(th);
  });
  thead.appendChild(headRow);
  table.appendChild(thead);

  const tbody = document.createElement("tbody");

  catStats.forEach(cat => {
    const tr = document.createElement("tr");
    tr.style.cssText = "background:#f5f5f5;border-bottom:1px solid #ccc;cursor:pointer";

    const nameCell = makeCell("td", "", { bold: true, title: cat.name });
    const arrow = document.createElement("span");
    arrow.dataset.arrow = cat.name;
    arrow.textContent = "▶";
    nameCell.appendChild(arrow);
    nameCell.append(" " + cat.name);
    tr.appendChild(nameCell);

    tr.appendChild(makeCell("td", String(cat.total), { center: true, bold: true }));
    tr.appendChild(makeCell("td", String(cat.on_cdse), { center: true, bold: true }));
    tr.appendChild(makeCell("td", badge(cat.s3, cat.total), { center: true }));
    tr.appendChild(makeCell("td", badge(cat.stac, cat.total), { center: true }));
    tr.appendChild(makeCell("td", badge(cat.openeo, cat.total), { center: true }));
    tr.appendChild(makeCell("td", badge(cat.shub, cat.total), { center: true }));
    tr.appendChild(makeCell("td", pctBar(cat.pct), {}));

    tr.addEventListener("click", () => window.toggleCategory(cat.name));
    tbody.appendChild(tr);

    cat.groups.forEach(g => {
      const groupKey = cat.name + "::" + g.name;
      const gtr = document.createElement("tr");
      gtr.dataset.category = cat.name;
```

```
      gtr.style.cssText = "border-bottom:1px solid #ddd;display:none;cursor:pointer";

      const gNameCell = makeCell("td", "", { title: g.name, padding: "4px 6px 4px 24px" });
      const hasDatasets = g.datasets && g.datasets.length > 0;
      if (hasDatasets) {
        const gArrow = document.createElement("span");
        gArrow.dataset.groupArrow = groupKey;
        gArrow.textContent = "▶";
        gArrow.style.marginRight = "4px";
        gNameCell.appendChild(gArrow);
      }
      const link = document.createElement("a");
      link.href = g.url;
      link.target = "_blank";
      link.textContent = g.name;
      link.addEventListener("click", (e) => e.stopPropagation());
      gNameCell.appendChild(link);
      gtr.appendChild(gNameCell);

      gtr.appendChild(makeCell("td", String(g.total), { center: true, bold: true, padding: "4px
6px" }));
      gtr.appendChild(makeCell("td", String(g.on_cdse), { center: true, bold: true, padding:
"4px 6px" }));
      gtr.appendChild(makeCell("td", badge(g.s3, g.total), { center: true, padding: "4px 6px" }));
      gtr.appendChild(makeCell("td", badge(g.stac, g.total), { center: true, padding: "4px 6px" }));
      gtr.appendChild(makeCell("td", badge(g.openeo, g.total), { center: true, padding: "4px
6px" }));
      gtr.appendChild(makeCell("td", badge(g.shub, g.total), { center: true, padding: "4px 6px" }));
      gtr.appendChild(makeCell("td", pctBar(g.pct), { padding: "4px 6px" }));

      if (hasDatasets) {
        gtr.addEventListener("click", () => window.toggleGroup(groupKey));
      }
      tbody.appendChild(gtr);

      (g.datasets || []).forEach(ds => {
        const dtr = document.createElement("tr");
        dtr.dataset.category = cat.name;
        dtr.dataset.group = groupKey;
        dtr.style.cssText = "border-bottom:1px solid #eee;display:none;background:#fafafa";

        dtr.appendChild(makeCell("td", ds.id, { padding: "3px 6px 3px 44px", title: ds.id }));
        dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));
        dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));
        dtr.appendChild(makeCell("td", check(ds.s3), { center: true, padding: "3px 6px" }));
        dtr.appendChild(makeCell("td", check(ds.stac), { center: true, padding: "3px 6px" }));
        dtr.appendChild(makeCell("td", check(ds.openeo), { center: true, padding: "3px 6px" }));
        dtr.appendChild(makeCell("td", check(ds.shub), { center: true, padding: "3px 6px" }));
        dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));

        tbody.appendChild(dtr);
      });
    });
  });
```

```
  // ── Summary row (col sums over the currently filtered groups) ──
  const sumRow = document.createElement("tr");
  sumRow.style.cssText = "background:#e0e0e0;border-top:2px solid #999;font-weight:bold";
  const sumName = makeCell("td", "Total", { bold: true });
  sumName.style.fontStyle = "italic";
  sumRow.appendChild(sumName);

  const sumTotal = d3.sum(catStats, d => d.total);
  const sumCDSE = d3.sum(catStats, d => d.on_cdse);
  const sumS3 = d3.sum(catStats, d => d.s3);
  const sumSTAC = d3.sum(catStats, d => d.stac);
  const sumOEO = d3.sum(catStats, d => d.openeo);
  const sumSHub = d3.sum(catStats, d => d.shub);
  const sumPct = Math.round(sumCDSE / Math.max(sumTotal, 1) * 100);

  sumRow.appendChild(makeCell("td", String(sumTotal), { center: true, bold: true }));
  sumRow.appendChild(makeCell("td", String(sumCDSE), { center: true, bold: true }));
  sumRow.appendChild(makeCell("td", badge(sumS3, sumTotal), { center: true }));
  sumRow.appendChild(makeCell("td", badge(sumSTAC, sumTotal), { center: true }));
  sumRow.appendChild(makeCell("td", badge(sumOEO, sumTotal), { center: true }));
  sumRow.appendChild(makeCell("td", badge(sumSHub, sumTotal), { center: true }));
  sumRow.appendChild(makeCell("td", pctBar(sumPct), {}));
  tbody.appendChild(sumRow);

  table.appendChild(tbody);
  wrap.appendChild(table);
  return wrap;
}

htl.html`
<div style="margin-bottom:8px;color:#888;font-size:0.9em">
  ${filtered.length} groups in ${catStats.length} categories
</div>
${productGroupsTable}
`
```

4. Services

- S3 CSV Catalogue
- OData API
- STAC Browser
- OpenEO
- Sentinel Hub
- CDSE Documentation

5. Methodology

Per-service counts are matched by:

- **S3**: Direct catalogue count per product type

- **STAC**: Fuzz-matched by checking if CDSE dataset ID (minus `_cog`/`_nc` suffix) appears in STAC collection IDs
- **OpenEO**: Matched by constructing `CLMS_{id.upper()}` convention
- **Sentinel Hub**: Estimated from S3 availability